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SET 1

MID SEMESTER EXAMINATION Mar-2025

Name of The Program : B.Tech(CE)  
Name OF the Course : Reinforced cement concrete-II  
Time: 90 Min

Semester : VI  
Course Code : TCE 602  
Max Marks : 50

Note: i) Answer **all the questions** by choosing any one of the sub questions.  
ii) Each question carries 10 Marks.

Q1 Answer any one.

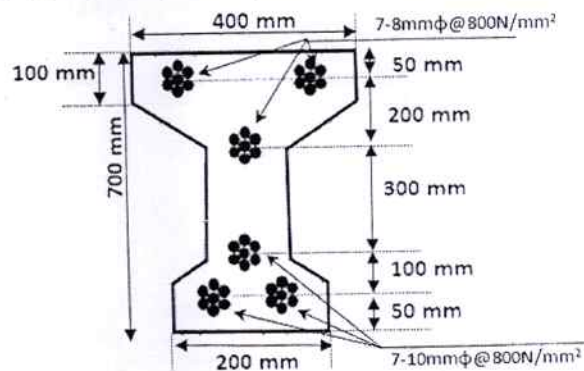
(1X10=10 Marks)

- (a) Compare the conceptual idea between RCC and PSC philosophy? (CO1)  
(OR)  
(b) Differentiate the techniques in any 5 post tensioning methods with figures? (CO1)

Q2 Answer any one.

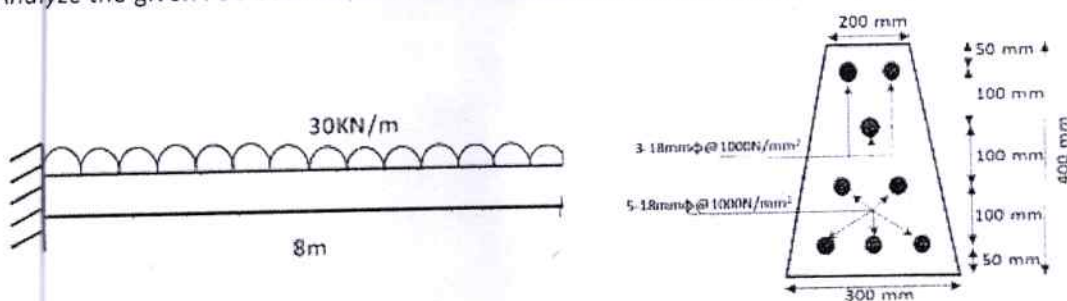
(1X10=10 Marks)

- (a) Determine the eccentricity of the PS Force for the given PSC beam section? (CO1)



(OR)

- (b) Analyze the given PSC beam section for maximum extreme fiber stresses? (CO1)



Q3 Answer any one.

(1X10=10 Marks)

- (a) Give a comparative discussion on three methods of analysis of a prestressed member? (CO1)

(OR)

- (b) Review the possible reasonings behind the IS Code recommendations for moment redistribution? (CO2)

Q4 Answer any one.

(1X10=10 Marks)

- (a) Illustrate the benefit of applying the limit analysis concept on a *Fixed Beam* carrying UDL over the entire span? (co2)

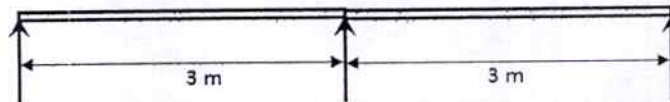
(OR)

- (b) Discuss the steps and observations in developing a bending moment envelop for a statically indeterminate beam? (co2)

Q5 Answer any one.

(1X10=10 Marks)

- (a) Develop the bending moment envelop after 20% moment redistribution for the given 2 span continuous beam for maximum negative bending moment at interior support? (co2)  
Self-weight of Beam: 1.5 kN/m, Dead Load on beam: 2.5 kN/m Live Load on Beam: 6kN/m



(OR)

- (b) Estimate the shift of point of contraflexure in the given fixed beam after applying 20% moment redistribution? (co2)

